

Q2: End-to-End Vendor Procurement System

QOSMIC SPACE — Founder's Office Interview

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1. Internal Purchase Request Form

A standardised intake form that captures everything the procurement person needs to source a part. Every field answers a specific downstream question.

Requester Information

Field	Required	Purpose
Requester name	Yes	Who to contact for clarifications
Department/Team	Yes	Optics, Mechanical, Electronics, Systems
Date requested	Yes	Start of procurement clock
Required-by date	Yes	Drives vendor selection (fast vs. standard lead time)
Budget code	Yes	Cost allocation and spend tracking
Cost centre manager	Yes	Approval routing

Part Information

Field	Required	Purpose
Part name	Yes	Human-readable identifier
Part number	Conditional	For reorders: previous PO or vendor P/N
New part or reorder	Yes	Reorders skip RFQ and spec review
Quantity	Yes	Total units needed
Estimated unit cost	Yes	Enables budget check before approval routing
Total estimated cost	Yes	Quantity × estimated unit cost

Specifications

Field	Required	Purpose
Part type	Yes	Dropdown: Catalog, Custom-machined, Custom-fabricated, COTS assembly
Drawing link	Conditional	Required for custom-machined parts
Material specification	Conditional	Required for custom parts: alloy/grade, temper
	Conditional	Critical dimensions the vendor must hold

Key dimensions/ tolerances		
Surface finish	Conditional	Ra value. Required for custom-machined.
Design document link	No	CAD assembly or system-level drawing

Abridged form for reorders: When the part has been procured within 6 months, only requester, department, date, quantity, required-by date, budget code, and part number are required.

2. Custom-Machined Parts Pre-Submission Checklist

Catches common specification errors before the request reaches a vendor. Items in Drawing and Material sections are hard gates — submission is blocked if any fail.

Drawing Requirements (Hard Gate)

1. **Drawing file attached and viewable** — PDF or DXF/STEP. No hand-drawn sketches.
2. **Fully dimensioned** — every feature has at least one dimension callout.
3. **Tolerance callouts on critical dimensions** — bilateral ($\pm 0.1\text{mm}$), unilateral, or GD&T.
4. **Complete thread callouts** — size $\times$ pitch $\times$ depth. The #1 rework cause.
5. **Revision letter and title block** — part name, material, scale, units, drawn by, date.
6. **Units explicitly stated** — mm or inches. Missing units = ₹5K-50K mistake.

Material Requirements (Hard Gate)

1. **Alloy/grade specified** — 6061-T6, not "aluminium."
2. **Temper specified** — T6, solution-treated, or "No HT required."
3. **Surface treatment specified** — clear anodise MIL-A-8625 Type II, black oxide, none.
4. **Consistency between drawing and form** — mismatched specs cause vendor confusion.

Tolerance & Fit

1. Mating features have explicit tolerance classes (clearance, transition, interference).
2. Surface finish on functional surfaces: $R_a \leq 0.8\text{ }\mu\text{m}$ optical faces, $R_a 3.2\text{ }\mu\text{m}$ non-functional.
3. Edge break / chamfer specified.
4. Flatness/parallelism for parts mating to optical breadboards.

Practical Checks (Warning — does not block submission)

1. Internal corners have relief radii matching available tool sizes.
2. Wall thickness $\geq 1\text{mm}$ (aluminium) or $\geq 0.5\text{mm}$ (steel).
3. Hole depths $\leq 4\times$ diameter for standard drills.
4. Thread depth $\leq 2.5\times$ thread diameter in aluminium.
5. Prototype history captured before production run.

Vendor-Specific Flags

1. Tight tolerances ($\pm 0.02\text{mm}$): verify vendor CNC capability.
2. Vacuum environment: outgassing checked for material/surface treatment.

3. Approval Thresholds & Justification

QOSMIC is a seed-stage startup at approximately ₹18L/month burn (~\$18,650/month at ₹96.5/USD). Cash is the constraint. Oversight scales with financial impact.

Purchase Total	Approval Required	Rationale
< ₹5,000 (< ~\$52)	Self-approved	0.28% of monthly burn. Wrong decision costs less than review overhead.
₹5,000 – ₹50,000 (\$52 – \$518)	CTO approval	2.8% of monthly burn. CTO assesses spec and price.
> ₹50,000 (> ~\$518)	CEO approval	Major optics, long-lead items. Strategic commitment.

Why ₹5,000: A ₹1,500 pack of bolts needing CTO sign-off costs 2×30 minutes of engineering time (≈₹2,000 in salary) for a ₹1,500 purchase. Self-approval is cheaper.

Why ₹50,000: One bad ₹50K order doesn't hurt. Three ₹50K orders in a month (₹1.5L) is 8.3% of monthly burn. CEO visibility prevents pile-up.

Emergency exception: Critical-path items where delay costs more than the part. CTO can approve without CEO bypass but must document the expedited reason.

Custom optical components (coated mirrors, precision lenses) always require CEO approval — 6-12 week lead times, cannot be returned.

4. Vendor Database Schema

Field	Type	Example
vendor_id	Auto-increment	V-2026-0042
company_name	String (required)	JLCPCB Co., Ltd.
primary_contact	String	Wei Zhang
contact_phone	Phone	+86-755-2394-1122
contact_email	Email	wei.zhang@jlcpcb.com
vendor_type	Enum	prototyping, production, local_fabrication, specialty
country	String	CN
capabilities	Multiple select	CNC, precision_optics, electronics, coatings, assembly
on_time_delivery_rate	Percent	94.5
quality_rejection_rate	Percent	2.1
avg_lead_time_days	Integer	18
payment_terms	Enum	net_30, net_45, LC, T/T
moq_standard_parts	Integer	50
moq_custom_parts	Integer	5
qualification_status	Enum	approved, conditional, new, inactive
last_interaction_date	Date	2026-04-12
total_spend_to_date	Currency (USD)	48,750.00

Sample Vendor Data (Airtable Screenshot)

	Company Name	Vendor ID	Primary Contact	Contact Phone	Contact Email	Vendor Type	Country	Capabilities
1	JLCPCB Co., Ltd.	V-2026-0001	Wei Zhang	+86-755-2394-1122	wei.zhang@jlcpcb.com	prototyping	CN	electronics, assembly
2	Jenoptik AG	V-2026-0014	Dr. Markus Reinhardt	+49-3641-65-2201	markus.reinhardt@jenoptik...	production	DE	precision_optics, c
3	Precision Bengaluru Works ...	V-2026-0027	Arvind Krishnamurthy	+91-80-4123-8800	arvind@pbw-machine.com	local_fabrication	IN	CNC, assembly
4	Thorlabs Inc.	V-2026-0008	Sarah Mitchell	+1-973-579-7227	s.mitchell@thorlabs.com	specialty	US	precision_optics, e

Figure 1: Vendors table in Airtable — JLCPCB, Jenoptik AG, Precision Bengaluru Works, Thorlabs Inc.

5. RFQ Process, Templates & Fast-Track Path

Vendor Solicitation Guidelines

Part Type	Vendors	Rationale
Catalog component	1-2	Published pricing. Second for lead time check.
Custom-machined part	3-5	Competitive. More than 5 = diminishing returns.
Precision custom optics	2-3	Fewer qualified vendors. Compare capability.
Electronics / PCBA	2-3	JLCPCB for prototype + local for production.

Fast-Track Path

For catalog components with a published part number and no customization: engineer specifies the vendor P/N, procurement goes directly to PO. No RFQ. Only for orders <₹50K and known vendors.

Template A: Initial RFQ Email

Subject: RFQ: Custom Aluminum Optical Mount — QOSMIC-OPT-2026-0012

Covers: part description (kinematic mount, 1-inch optic, 1064nm), material (6061-T6, clear anodised), quantity (5/25/50 tiers), tolerances (Ra 0.8 µm, ±0.05mm), drawing reference, delivery (first article 4 weeks), shipping terms (FCA), and explicit request for cost breakdown.

Template B: Follow-Up (5 business days of silence)

Polite nudge. "If you need more time, a quick note helps. If unable to quote, a brief reply is fine."

Template C: Negotiation (quote 30%+ above budget)

Firm but constructive. Asks for cost breakdown. Offers specific concessions: switch anodise to as-machined on non-functional surfaces, relax Ra on non-critical faces, accept longer lead time. "If X% off is achievable, we can issue the PO today."

6. PO Tracking System

Status Flow

Issued → Acknowledged → In Production → Shipped → In Transit → Delivered → Inspected → Accepted / Rejected

↓
Customs Hold

Automated Alert Triggers

Trigger	Condition
PO not acknowledged	Status = Issued for >3 business days
Ship date approaching	5 days before promised ship, no confirmation
Delivery overdue	Past promised delivery date
No tracking update	In Transit for >7 days with no location update
Customs hold >3 days	On expensive or time-sensitive items
Quarterly budget exceeded	Vendor spend >120% of quarterly budget
Inspection pending	Delivered for >2 business days

International Shipment Tracking

Field	Example
HS Code	9013.80 (optical instruments)
CIF Value	\$1,244
BCD Rate	7.5% (verified: cybex.in)
IGST Rate	18% (standard for Chapter 90)
Effective Duty Rate	~27.7% (HS 9013) / ~31% (HS 9002)
Customs Broker Fee	₹2,000-10,000 (verified: azafra.mx)
Shipping Insurance	0.5-2% of CIF (required for >\$518)

Kanban Board Screenshots

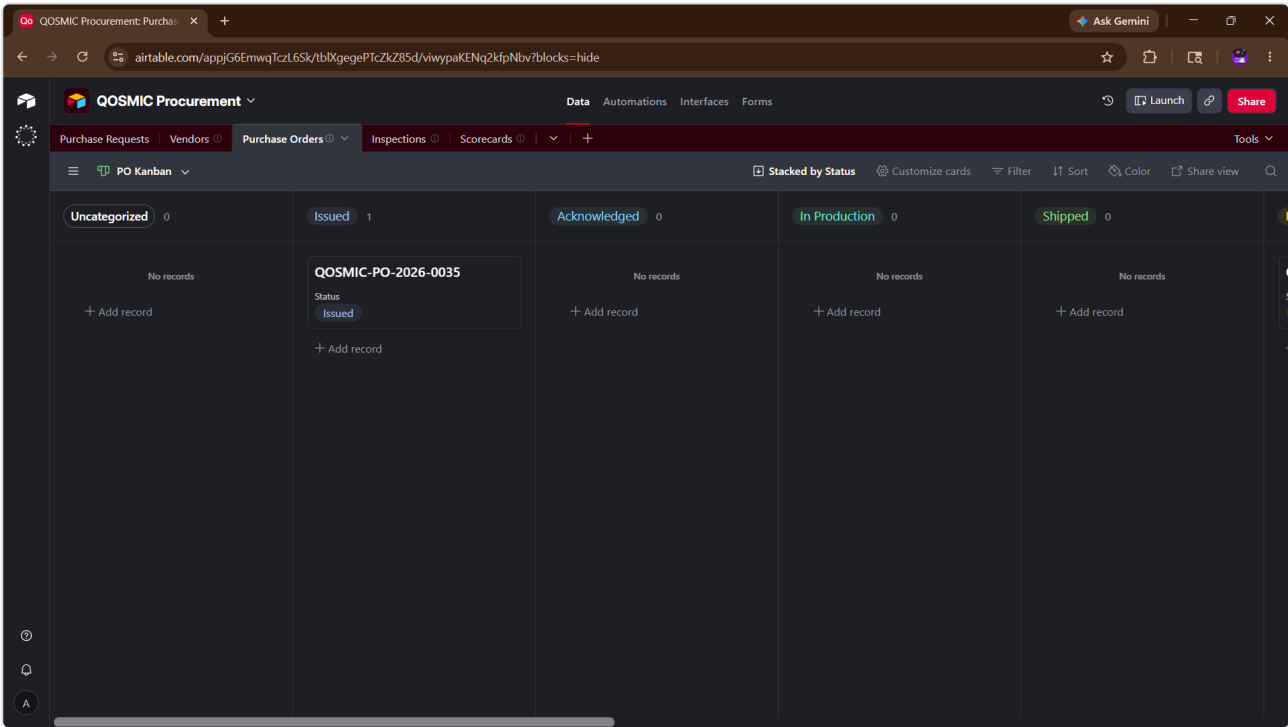


Figure 2a: PO Kanban — Issued column with PO-2026-0035 (M6 fasteners, self-approved)

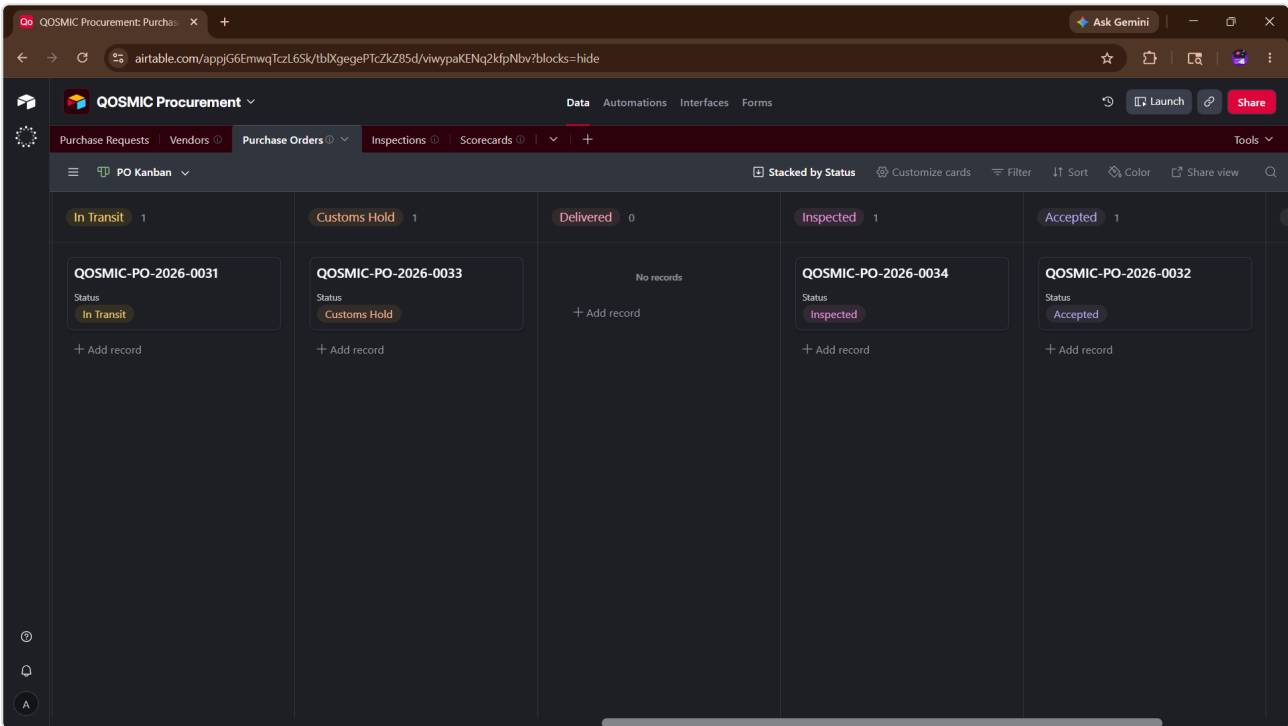


Figure 2b: PO Kanban — In Transit (PO-0031), Customs Hold (PO-0033), Inspected (PO-0034), Accepted (PO-0032)

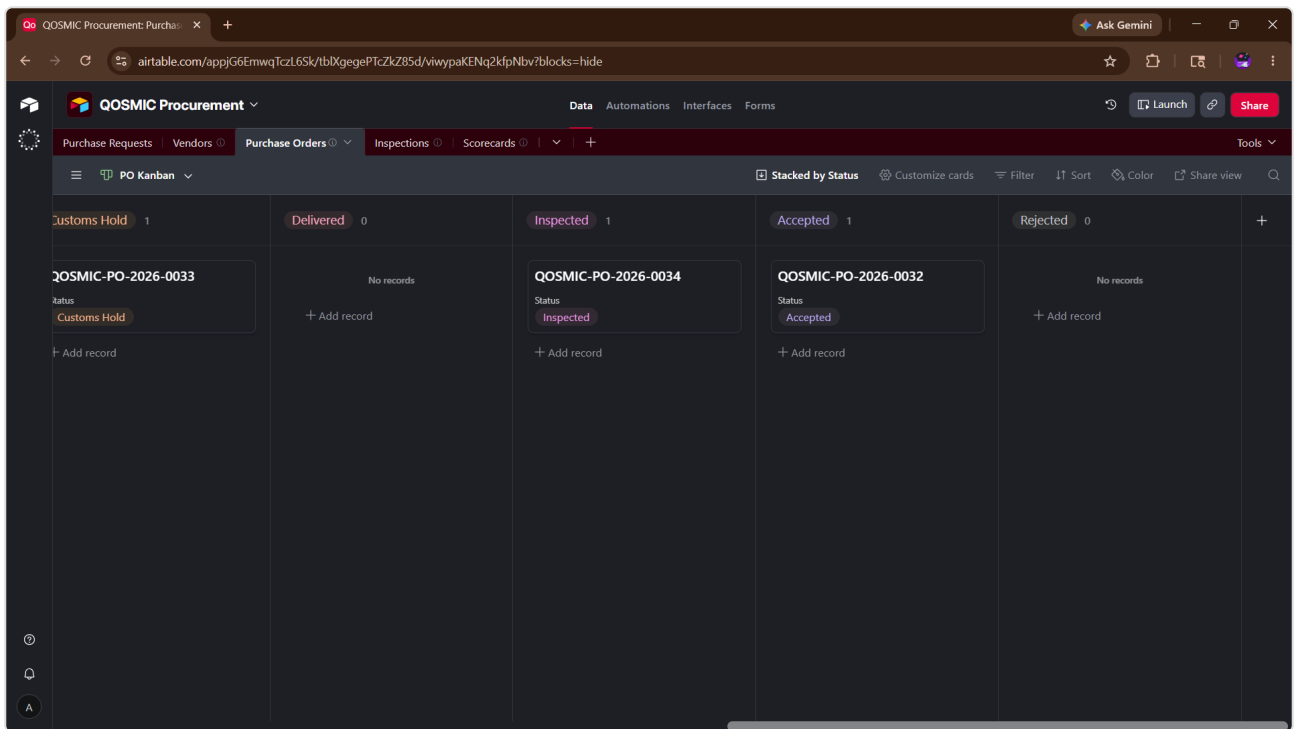


Figure 2c: PO Kanban — Customs Hold column with PO-2026-0033 (custom Jenoptik lens, stuck at Bengaluru customs)

7. Incoming Inspection Checklists

Custom-Machined Parts

Critical (must pass): Quantity match, visual defects, dimensional check of mounting surfaces, thread gauge check, surface finish on optical faces ($Ra \leq 0.8 \mu m$), material certification.

Warning (conditional accept): Edge break/chamfer dimensions, colour/anodise consistency, burrs on tapped holes, packaging quality.

Precision Optical Components

Parameter	Why It Matters	Threshold
Surface quality (scratch-dig)	Scratches scatter 1064nm light, reduce SNR, initiate laser damage	20-10 (MIL-PRF-13830B)
Wavefront error (WFE)	10 mirrors at $\lambda/10$ each = 1λ total. Fibre coupling drops 80% $\rightarrow$ 30%.	$\lambda/10$ @ 633nm
Coating reflectivity (HR)	99.8% R = 0.2% loss/surface. 15 surfaces = 3% vs. 14% loss at 99.0%.	$\geq 99.5\%$ @ 1064nm
Clear aperture	Beam clipping creates diffraction rings	$\geq 90\%$ of physical diameter
Centration (lenses)	0.1mm decentration = milliradian beam misalignment	$\leq 0.05mm$

Seed-stage reality: QOSMIC won't own an interferometer. Rely on vendor test reports, \$165 scratch-dig paddle (Edmund Optics), and ₹5K-15K independent verification at IISc/ARCI for first articles.

8. Rejection Workflow & Partial Shipments

Workflow

Inspection finds defect → Photograph (3+ angles, scale reference) → Move to quarantine → Send rejection email → 3 business day response window → Resolution (rework / replace / refund) or escalate to CEO.

Rejection Email (Template D)

Specifies measured vs. specified values in a table, attaches photographs and inspection report, and requests one of three resolutions within 3 business days.

Partial Shipment Policy

- Partial shipments allowed for multi-line POs or quantity >10.
- Each partial gets its own receiving record and inspection.
- Accepted units paid per PO terms; rejected units follow rejection workflow.
- Track rejection rate by unit count, not dollar value — prevents expensive single parts from masking systemic quality issues.

9. Vendor Quality Scorecard

Metric	Target	Weight
On-time delivery rate	≥ 90%	25%
Quality rejection rate	≤ 3%	25%
Quote accuracy (deviation from final invoice)	≤ 5%	10%
Lead time consistency (std dev from quoted)	≤ 7 days	10%
Responsiveness (avg time to respond)	≤ 2 business days	10%
Documentation quality (% complete)	100%	10%
Issue resolution speed	≤ 5 business days	10%

Scoring: Each metric 0-100. Composite = $\Sigma(\text{score} \times \text{weight})$. ≥85 = Preferred, 70-84 = Approved, 55-69 = Conditional, <55 = Probation.

Scorecard Screenshot

Scorecard ID	Documentation Score	Issue Resolution...	Composite Score	Status	Notes	Vendor
1	100	85	89	Preferred		Jenoptik AG
2	75	60	66	Conditional		Precision Bengaluru Works Pvt. Ltd.
3	100	95	94	Preferred		JLCPCB Co., Ltd.
4	100	95	97	Preferred		Thorlabs Inc.
Sum	375	335	346			

Figure 3: Scorecards table — Jenoptik 89 (Preferred), PBW 66 (Conditional), JLCPCB 94 (Preferred), Thorlabs 97 (Preferred)

10. Tool Stack Proposal

Tool	Plan	Cost	Role
Airtable	Free	\$0	Core database: forms, records, Kanban, automations
Gmail	Free	\$0	Email for vendor communication
Slack	Free	\$0	Alert channel (#procurement-alerts)
Make.com	Free	\$0	Connect Airtable → Slack → Gmail
Google Drive	Free	\$0	File storage
Total tooling cost		\$0	

Why Airtable over alternatives: Notion's automations are too weak for conditional routing. Excel has no relational integrity or automations. Custom web app violates the "usable by non-developer" constraint. Airtable is the only no-code tool combining forms, relational databases, automations, and Kanban/Calendar/Form views in one platform.

Scaling path: At 200+ POs/month (years away), total cost reaches \$66/month — well under the \$200 ceiling.

11. AI/LLM Automation Opportunities

Task	Level	Implementation
Form completeness checking	Fully automatable	Airtable form validation (rule-based)
Approval routing	Fully automatable	If/then rules based on total cost
PO status tracking	Fully automatable	Airtable Kanban + Make.com alerts
Scorecard calculation	Fully automatable	Airtable formula fields
RFQ email drafting	Partially automatable	LLM fills template, human reviews
Follow-up email drafting	Partially automatable	LLM per tone guidance, human reviews
Quote PDF parsing	Partially automatable	LLM extracts into comparison table
Customs doc pre-fill	Partially automatable	LLM pre-fills, human verifies HS code
Scorecard narrative	Partially automatable	LLM generates paragraph, human edits
Rejection email drafting	Partially automatable	LLM formats, human verifies measurements
Vendor selection decision	Human-judgment	Strategic — workload, relationships, fit
Approval decisions	Human-judgment	CTO/CEO must see spec and cost
Incoming inspection	Human-judgment	Physical measurement cannot be automated
Negotiation	Human-judgment	Reading the vendor's situation

Priority: Quote PDF parsing first (saves 3-5 hrs/month), then RFQ drafting (1-2 hrs), then follow-ups (1-2 hrs). Total AI cost: ~\$0.50/month in Claude API calls.

12. Airtable Prototype — 5 Sample POs

Base Structure

- **Table 1: Purchase Requests** — Form intake, linked to POs
- **Table 2: Vendors** — Vendor database, 17 fields
- **Table 3: Purchase Orders** — 10-status Kanban, linked to vendors
- **Table 4: Inspections** — Results linked to POs
- **Table 5: Scorecards** — 7 weighted metrics, auto-calculated

Sample PO Data

PO#	Part	Vendor	Status	Value	Scenario
2026-0031	Custom aluminum mounts	PBW	In Transit	₹35,000	Shipped 2 days late
2026-0032	SM1 lens tubes	Thorlabs	Accepted	₹18,500	Fast-track. Arrived early.
2026-0033	Custom aspheric lens	Jenoptik	Customs Hold	₹1,20,000	Stuck at customs. CEO notified.
2026-0034	PCB assembly	JLCPCB	Inspected	₹8,200	9/10 passed. 1 rejected.
2026-0035	M6 fasteners	PBW	Issued	₹2,800	Self-approved reorder.

PO Record Screenshot

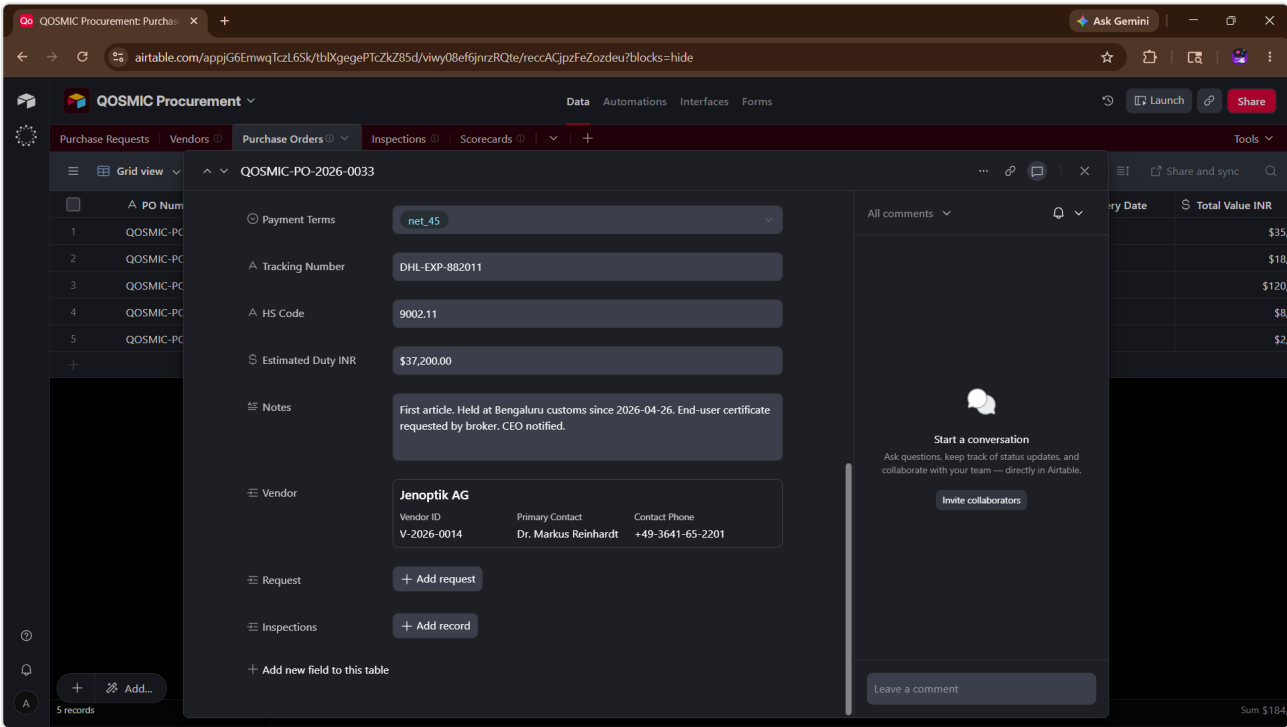


Figure 4: PO record expanded view — linked Vendor field, status dropdown, and tracking fields visible

13. Implementation Cost Estimate

One-Time Setup

Item	Cost
Airtable base setup (5 tables, views, links)	\$0 (self-built, free plan)
Template documents	\$0 (created and included in this submission)
Scratch-dig paddle (Edmund Optics #91-291)	\$165 (only if doing in-house optical QC)
Digital calliper (0.01mm)	\$20-50
Total one-time	\$0-215

Recurring Monthly

Item	Cost
Airtable Free	\$0
Make.com Free	\$0
Slack Free	\$0
Gmail	\$0
Claude API (AI automations)	~\$0.50
Customs broker fee (per international shipment)	\$31-83
Total tooling	\$0-6.50/month

The \$200/month budget constraint is not approached at any realistic scale — even at 200+ POs/month, total tooling is ~\$66/month.